



Regional soil organic carbon prediction model based on a discrete wavelet analysis of hyperspectral satellite data

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ABSTRACT

Most studies have achieved rapid and accurate determination of soil organic carbon (SOC) using laboratory spectroscopy; however, it remains difficult to map the spatial distribution of SOC. To predict and map SOC at a regional scale, we obtained fourteen hyperspectral images from the Gaofen-5 (GF-5) satellite and decomposed and reconstructed the original reflectance (OR) and the first derivative reflectance (FDR) using discrete wavelet transform (DWT) at different scales. At these different scales, as inputs, we selected the 3 optimal bands with the highest weight coefficient using principal component analysis and chose the normalized difference index (NDI), ratio index (RI) and difference index (DI) with the strongest correlation with the SOC content using a contour map method. These inputs were then used to build regional-scale SOC prediction models using random forest (RF), support vector machine (SVM) and back-propagation neural network (BPNN) algorithms. The results indicated that: 1) at a low decomposition scale, DWT can effectively eliminate the noise in satellite hyperspectral data, and the FDR combined with DWT can improve the SOC prediction accuracy significantly; 2) the method of selecting inputs using principal component analysis and a contour map can eliminate the redundancy of hyperspectral data while retaining the physical meaning of the inputs. For the model with the highest prediction accuracy, the inputs were all derived from the wavelength range of SOC variations; 3) the differences in prediction accuracy among the different prediction models are small; and 4) the SOC prediction accuracy using hyperspectral satellite data is greatly improved compared with that of previous SOC prediction studies using multispectral satellite data. This study provides a highly robust and accurate method for predicting and mapping regional SOC contents.

1. Introduction

Soil is the largest terrestrial carbon reservoir in the biosphere, accounting for approximately 75 % of the total carbon pool in terrestrial ecosystems. Slight changes in carbon reserves may lead to significant differences in the atmospheric concentration of CO₂, thus affecting the global climate (Luo et al., 2010). Accordingly, the monitoring and rapid digital mapping of soil organic carbon (SOC) are among the most important endeavours for developing strategies to mitigate global warming (Jones et al., 2005). To date, most SOC measurements have been conducted in labs, and collecting soil samples is usually time

consuming and destructive. Moreover, discrete soil samples cannot provide continuous information regarding the spatial characteristics of soil properties; thus, it is difficult to map SOC at regional and global scales (Taghizadeh-Mehrjardi et al., 2016). Rapidly quantifying the spatial distribution of SOC content has become a research focus, and remote sensing techniques and machine learning algorithms have become powerful assessment tools (Camera et al., 2017). Compared with laboratory SOC prediction approaches, methods using remote sensing data in the visible, near-infrared and shortwave infrared (VNIR/SWIR, 0.4–2.5 μm) domains have been found to be faster and more cost efficient (Chang and Laird, 2002).

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Previous studies have demonstrated the feasibility of SOC prediction using imaging spectroscopy data acquired by different sensors. The statistical relationships between the SOC content and the intensity values of the red, green and blue bands of aerial photographs were studied by [Chen et al. \(2000\)](#) at a small-scale, and an exponential regression model was established. To predict soil properties, [Castaldi et al. \(2016\)](#) evaluated the performance of data from different satellites, including the Earth Observing (EO)-1 Advanced Land Imager (ALI), Hyperion imaging spectrometer, Landsat 8 Operational Land Imager (OLI), and Sentinel-2 Multispectral Instrument (MSI), as well as the forthcoming Environmental Mapping and Analysis Program (EnMAP) mission, PRISMA instrument and Hyperspectral Infrared Imager (HypIRI). Their study demonstrated that the SOC prediction accuracies using Hyperion data and multispectral data (such as Landsat 8 OLI and Sentinel-2 MSI), are comparable because of their low spectral resolutions and low signal-to-noise ratio (SNR), whereas forthcoming hyperspectral imagers are expected to have better prediction accuracy. Notably, the rich information provided by these bands of the hyperspectral imagers helps capture details related to such predictions, especially the prediction of soil properties; it should be noted, however, that single-date spectral information is more abundant than multitemporal multispectral remote sensing information ([Dou et al., 2019](#)).

Most hyperspectral data used for assessing environmental variables are acquired by airborne or unmanned airborne spectral imagers ([Suomalainen et al., 2014](#)). [Paz-Kagan et al. \(2015\)](#) evaluated the spectral soil quality index in a 144 ha experimental area using AISA FENIX (380–2500 nm, 448 spectral bands) and AISA DUAL (420–2450 nm, 367 spectral bands) airborne data with a partial least squares regression (PLSR) prediction model. The correlation coefficient between the spectral soil quality index and the derived soil quality indexes was 0.71, suggesting that the soil environmental quality can be effectively monitored using spectral information. However, drones are expensive and susceptible to atmospheric disturbances during their operation; furthermore, these sensors have a limited number of spectral bands and a low SNR ([Ben-Dor et al., 2009](#)). In addition, airborne sensors cannot satisfy the needs for rapid mapping at the regional to global scales, whereas hyperspectral airborne imagery can be used as an important information source for large-scale soil mapping ([Lagacherie and McBratney, 2006](#)). Satellite hyperspectral sensors are designed to acquire data with spectral resolutions of 20 nm or less ([Table 1](#)) ([Mirik et al., 2005](#)) and obtain spectral information over a large area in a short

time ([Tahmasbian et al., 2018](#)). GF-5 satellite hyperspectral sensors can obtain full-spectrum observations from ultraviolet to longwave infrared spectra and detect the specific components of substances, and these sensors form the first high-precision "image-spectrum merging technology" spectroscopy analyser in the GF series. Compared with other hyperspectral satellite sensors, GF-5 contains a large amount of band information. The spectral resolution of the shortwave infrared band is 5 nm, up to 0.03 beams. GF-5 is the first integrated hyperspectral satellite for both atmospheric and land observations worldwide. [Table 1](#) lists the main technical specifications of various hyperspectral satellite systems.

Before the full-spectrum hyperspectral satellite GaoFen-5 (GF-5) was launched, Hyperion of the NASA EO-1 project and the Compact High Resolution Imaging Spectrometer (CHRIS) of the European Space Agency's Project for On-Board Autonomy (PROBA) were the two main space-based hyperspectral sensors ([Castaldi et al., 2016](#)). [Zhang et al. \(2013\)](#) conducted a study to quantitatively estimate soil properties (soil moisture, soil organic matter (SOM), total carbon (C), total phosphorus (P), total nitrogen (N) and the clay content) in the United States and Spain using Hyperion data. However, both Hyperion and CHRIS have certain shortcomings. For instance, the SWIR spectral channels (approximately 2200 nm) of Hyperion have a low SNR, and the sensor was officially decommissioned in 2017. CHRIS only covers the visible NIR spectral range (415–1050 nm) ([Casa et al., 2013](#)). Thus, both sensors have limitations for applications in predicting soil properties.

Constrained by a narrow bandwidth of spectral channels, hyperspectral satellite data often have a high level of noise ([Shafri and Yusof, 2009](#)); thus, the extraction of spectral features with enhanced information is an important step in building prediction models ([Vašát et al., 2017](#); [Siegmann et al., 2012](#)). Wavelet analysis is a method of spectral analysis that extracts spectral features. In particular, discrete wavelet transform (DWT) can be used to reduce high-frequency spectral noise while preserving the useful spectrum information for SOC prediction ([Viscarra and Behrens, 2010](#)). Derivative spectral analysis is another method of spectral analysis that removes background effects and enhances subtle spectral features. However, literature reviews have shown that different spectral processing methods have led to inconsistent results in different experiments, and the application capacity of remotely sensed spectral reflectance data for soil property estimation is still unclear. Further study of these topics is thus required.

Various prediction models have been built using machine learning

Table 1
Technical specifications of a few hyperspectral satellite systems.

Imager	Spectral bands	Spectral range (nm)	Spatial resolution	Spectral resolution	Time resolution	SNR	SNR condition
Hyperion	220	400–2500	30 m	10 nm	16 d	161 @ 550 nm 147 @ 700 nm 110 @ 1125 nm 40 @ 2125 nm	Nadir looking 60°, sun-zenith angle, 0.3 Earth albedo
CHRIS	37	400–1050	17/34 m	6–33 nm	3 d	1.3 nm @ 410 nm 12 nm @ 1050 nm	Solar synchronous orbit, altitude 615 km, obliquity 97.89°
HJ-1	128	459–956	100 m	5 nm	4 d	2–3 nm @ 460 nm 3–5 nm @ 552 nm 5–7 nm @ 716 nm 7–8 nm @ 848 nm	± 30° side swaying for global repetitive observations
MODIS	36	400–1400	250 m (1–2 bands) 500 m (3–7) 1000 m (8–36)	5–10 nm	1–2 d	128 @ 620–720 nm 201 @ 841–876 nm	Derailed transit at 10:30 am, transiting at 1:30 pm transit; sun synchronization; near polar circular orbit
FTHSI	256	475–1050	30 m	2–10 nm	20–30 min	military	Angle of view of 150°
GaoFen-5	330	400–2500	30 m	5 nm (400–1000 nm) 10 nm (1000–2500 nm)	7 d	The maximum SNR 450–2500 nm is 500	Sun-synchronous orbit with an inclination of 98.2°, orbital height is approximately 705 km

Note: The SNR column reports the signal-to-noise ratio values and the wavelength at which they are calculated. The specific meaning of @ refers to the position corresponding to the calculation of the SNR.

techniques. Nawar et al. (2016) showed that a model based on PLSR can perform well at predicting SOM, whereas Knox et al. (2015) found that a model based on the random forest (RF) approach is better, particularly for data that do not follow a normal distribution. Viscarra Rossel and Behrens (2010) reported that a model based on a support vector machine (SVM) can achieve more accurate predictions than PLSR and RF models. Xu et al. (2018) demonstrated that models based on back-propagation neural network (BPNN) and SVM methods provide better performance than principal component regression (PCR) and PLSR models in the prediction of soil properties. Many prediction models are site specific, showing good performance for the regions for which they are calibrated but not in other regions (Chang et al., 2001). Therefore, the selection of calibration methods and the ability of a model to adequately interpret spectral information are important for successfully performing predictions (Mouazen et al., 2010). Further research is needed to reduce noise in hyperspectral data, select and extract optimal spectral features as predictor variables and establish a highly accurate SOC estimation model. This study investigated the use of DWT analysis and first derivatives to reduce noise in hyperspectral remote sensing data acquired by the GF-5 satellite and established an SOC prediction model for application in the northern Songnen Plain region in Northeast China. The main objectives were as follows: 1) to develop a feature extraction method to reduce noise in hyperspectral data and improve SOC prediction, 2) to establish SOC estimation models using different approaches and compare their performance and 3) to map the spatial variability of SOC at a regional scale.

2. Materials and methods

2.1. Study area

Northeast China is the region with the largest amount of cultivated agricultural land in China. Dominated by annual crops, this region still utilizes a traditional residue management method and burns crop residue. The “bare soil period” refers to the period when there is little crop residue and no snow cover in fields. Remote sensing images acquired during this period are not affected by straw, green crops or snow; hence, this period is ideal for mapping the variability of soil properties. The “bare soil period” in the study area occurs from late March to late May, a period with good conditions for satellite image acquisition, and the cultivated land went into fallow state. Fig. 1c shows the surface conditions of the cultivated land within the study area during the bare soil period. The study area, located within the Songnen Plain in Northeast China, is located in the southern part of Heilongjiang Province (Fig. 1a) and has a temperate continental, semi-humid and semi-arid monsoon climate according to the Köppen-Geiger climate classification system. The area is warm and rainy in summer and cold and dry in winter. The main soil types in the region are

Phaeozems, Chernozems, Arenosols, and Cambisols, as described in the World Reference Base for Soil Resources (WRB). Arenosols are mainly distributed in the southwestern portion of the study area, with a low clay content and poor water-holding capacity, and they have a lower SOC content than the other soil classes. The soil types in the north-eastern portion of the region are mainly Phaeozems and Cambisols, which contain more humus and a higher SOC content than the other soil classes.

2.2. Soil sample collection and treatment

A total of 315 soil samples were collected from the study area, and 126, 64, 71 and 54 samples were collected in 2014, 2016, 2017 and 2018, respectively. With similar cropping practices throughout the study period (2014–2018), we assumed that changes in SOC were inconsequential and could be ignored. The sampling sites were distributed across the whole geographic extent of the study area and included all the main soil types (Phaeozems, Chernozems, Arenosols and Cambisols). These samples allow for the establishment of an SOC prediction model with high universality. The specific collection process was as follows. The sample location was selected according to a soil map and a cultivated land map. Then, 5–6 topsoil subsampling points (0–20 cm) were randomly selected in a square grid within an area of 30 by 30 m. Then, soil samples were mixed to obtain a mixed sample for each square grid. A portable global positioning system (GPS, G350, UniStrong, China) unit was employed to record the location and time of each sample collected. The soil was air dried, ground and sieved through a 2-mm sieve to obtain 5 g of soil, and the SOC content was measured using the potassium dichromate heating method (Nelson and Sommers, 1974).

2.3. Hyperspectral satellite data acquisition and treatment

2.3.1. Hyperspectral satellite data acquisition

Hyperspectral data were acquired by GF-5 satellite and obtained through the Chinese Earth Observation Satellite Data Service (<http://data.cresda.com:90/#/home>). During a “warm winter”, the images acquired in March and April of 2019 were used as the bare soil period images. Fourteen cloud-free images were available from early March to late April during this period. The specifications of the sensor are provided in Table 1. Namely, the spatial resolution is 30 m, the swath width is 60 km and the spectral range is from 390 to 2513 nm with a total of 330 spectral channels. The VNIR spectral range contains 150 spectral bands with a spectral resolution of 4.28 nm, and the SWIR range contains 180 spectral bands with a spectral resolution of 8.42 nm.

2.3.2. Data preprocessing

Images were radiometrically calibrated and then atmospherically

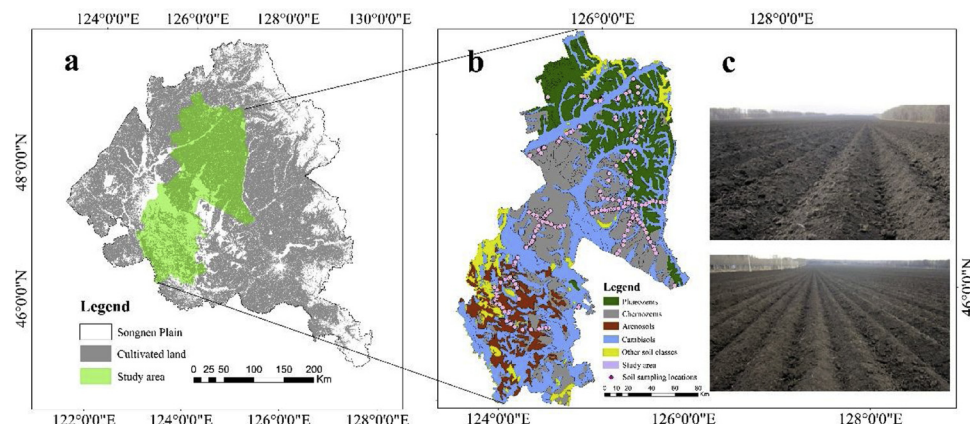


Fig. 1. Overview of the study area (a), soil classes and soil sampling locations (b), and (c) photographs of the soil surface during the bare soil period.

corrected to convert to surface reflectance using the FLAASH (Fast Line-of-sight Atmospheric Analysis of Spectral Hypercubes) module in ENVI 5.3 (Environment for Visualizing Images 5.3, Harris Geospatial Corporation, USA). The spectrum corresponding to each sampling point was extracted from the images. Because the sensor has a low SNR in the spectral ranges of 390–433 nm and 2445–2513 nm and the wavelengths of 1350–1451 nm and 1771–1982 nm are affected by atmospheric water vapor absorption, the spectral channels in these ranges were excluded from this study.

2.3.3. First derivative reflectance (FDR)

FDR processing can remove the background effects on plain spectral features, thus helping to reduce systematic errors in spectral data (Tsai and Philpot, 2002). When the contents of some components in the soil are low and spectral features are not strong, first-derivative processing can enhance the spectral features for SOC estimation (Tsai and Philpot, 1998).

2.3.4. Discrete wavelet transform (DWT)

Hyperspectral data can provide detailed spectral information about a target; however, the data are usually noisy (Shafri et al., 2009). DWT can be employed in analyses based on different spectral regions, and this method is especially suitable for spectrally localized noise reduction (Qian, 2005). DWT includes two processes, namely, signal decomposition and signal reconstruction. Decomposition is a process that decomposes the spectral signal into low-frequency and high-frequency components using a set of base wavelets. This procedure is often conducted iteratively to progressively lower the frequency or larger scale, until a maximum scale dependent on the signal length and the wavelet basis length is reached. Fig. 2 shows the process of using DWT to decompose an original hyperspectral reflectance spectrum (hyperspectral data) into low-frequency and high-frequency components. The low-frequency component consists of wavelet coefficients, where each coefficient is directly related to the amount of energy in the signal at different positions and scales, and the high-frequency component

mainly contains noise. At a decomposition scale, the low-frequency component A_j , and high-frequency component D_j are derived. Thus, the original hyperspectral signal S is the sum of the low-frequency component at the final scale J and high-frequency component at all scales $j = 1, \dots, J$ (Blackburn, 2007):

$$S = A_J + \sum_{j=1}^J D_j \quad (1)$$

The aim of the signal reconstruction process is to eliminate noise in the high-frequency component with a filter and reconstruct the low-frequency component of each scale. Thus, a visual representation of 'where' the best wavelet coefficients reside within the spectral curve can be obtained (Koger et al., 2003; Asim et al., 2011). In DWT, the mother wavelet function and the scale of decomposition are needed. In this paper, we tested a few mother wavelet functions in the MATLAB wavelet toolbox, including the Haar wavelet, Daubechies (dbN) wavelet, and Symlets (symN) wavelet. After repeated tests, we selected the 'db4' mother wavelet, and the decomposition scale was chosen such that it was maximized (on a scale of 10), as represented by L_1 - L_{10} ; L_0 represents a hyperspectral signal that was not processed by DWT analysis.

2.4. Selection of predictors for SOC estimation

2.4.1. Selection of the optimal bands

Currently, most studies based on principal component analysis (PCA) convert soil hyperspectral reflectance data into several independent principal components as the inputs for SOC prediction (Hong et al., 2019), but the principal components are difficult to physically interpret. For the reasons mentioned above, in this paper, we attempt to use the PCA method to analyse the soil hyperspectral reflectance bands, to calculate each band's weighting coefficient W and to extract the bands with the highest W values as the inputs of the prediction model instead of extracting the principal component information directly. This method not only preserves the meaning of the inputs but also eliminates

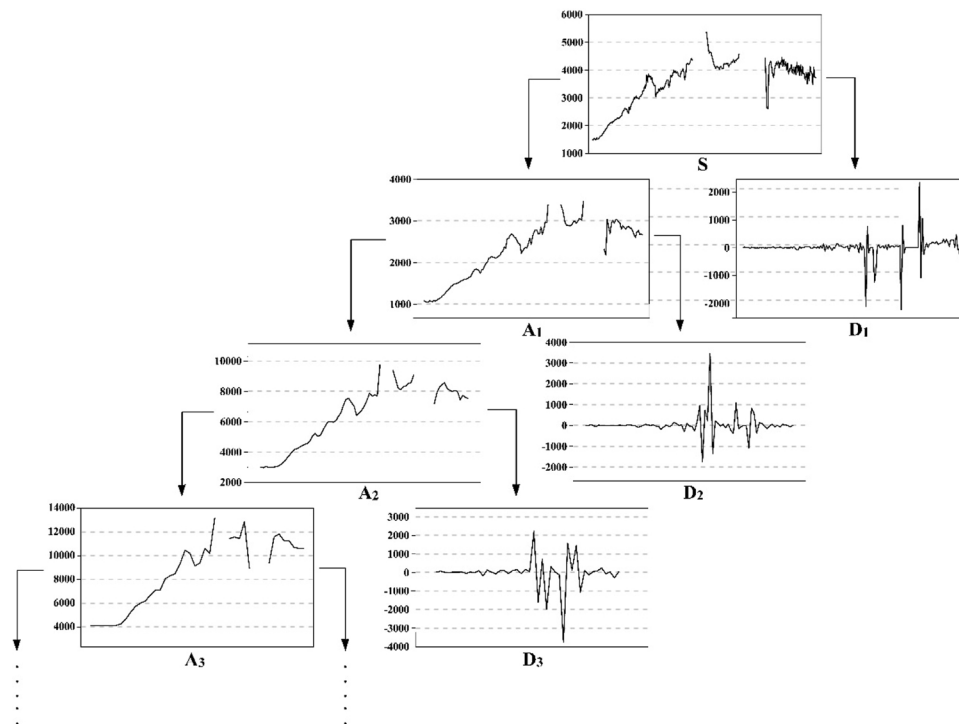


Fig. 2. DWT decomposition process.

Note: S is the hyperspectral curve; A_1 - A_3 denote the low-frequency components from decomposed scale₁ to decomposed scale₃; and D_1 - D_3 denote the high-frequency components from decomposed scale 1 to decomposed scale 3.

the high collinearity among bands. The specific calculation process of W is as follow:

First, we perform PCA on hyperspectral data and select the principal components which the explained amount of the total spectral variance is greater than 85 %, and calculate the α of each band in each principal component according to formula (2).

Last but not least, the W of each band are calculated according to formula (3).

$$\alpha_{ni} = \frac{\varepsilon_{ni}}{\sqrt{\sigma_{ni}}} \quad (2)$$

$$W_n = \frac{\alpha_{n1} \times \partial_{n1} + \alpha_{n2} \times \partial_{n2} + \dots + \alpha_{ni} \times \partial_{ni}}{\partial_{n1} + \partial_{n2} + \dots + \partial_{ni}} \quad (3)$$

where W_n is the weighting factor corresponding to band n and α_{ni} , σ_{ni} , ∂_{ni} , and ε_{ni} are the linear coefficient, initial eigenvalue, initial eigenvalue variance and load number corresponding to the i th principal component of the n th band respectively.

2.4.2. Selection of the optimal spectra index

Inoue et al. (2012) showed that the contour map method can be efficiently applied to select the optimal combination of bands for use in existing sensors and for designing future sensors, and the results can easily be compared with those of other studies. Contour maps clearly show the relationship between any combination of two wavelengths and soil properties. For example, the correlation (R) value between SOC and spectral parameters can provide comprehensive information regarding the relationships of combinations of separate wavelengths of hyperspectral data. In this study, we attempted to select the optimal spectral index by using the contour map method for satellite hyperspectral data. The normalized difference index (NDI), ratio index (RI) and difference index (DI) have proven useful in predicting the SOM content (Jin et al., 2016); thus, we applied the NDI, RI and DI equations in the contour map method and selected the optimal spectral index as a predictor in the SOC prediction model. The formulas for the NDI, RI and DI are as follows:

$$NDI(R_i, R_j) = (R_i - R_j) / (R_i + R_j) \quad (4)$$

$$RI(R_i, R_j) = R_i / R_j \quad (5)$$

$$DI(R_i, R_j) = R_i - R_j \quad (6)$$

where R_i and R_j are the reflectance values selected from the 433–1342, 1460–1763, and 1990–2445 nm bands and $i \neq j$.

2.5. Prediction models

In this paper, 6 inputs were selected. 3 bands with the highest W values and NDI, RI and DI values with the highest correlations with the SOC content were used as the inputs for the SOC prediction models established using the RF, SVM and BPNN algorithms for SOC content estimation.

2.5.1. Random forest (RF)

RF regression can be considered an ensemble regression approach involving multiple decision trees. For the training data set, the bootstrap resampling method is used to construct m decision trees using m samples. Each split node in the decision tree is randomly selected from n inputs such that the variable space can be completely divided. The average of these decision tree predictions is taken as the predicted dependent variable (Breiman, 2001). The RF algorithm not only provides fast training and prediction but also avoids overfitting because of its random sampling approach (Grimm et al., 2020). We used the randomForests package of Liaw and Wiener (2002) in R software (R Development Core Team, 2008) for the prediction. When training the RF algorithm, we tested different values for the tuning parameters, namely, the number of variables randomly sampled as candidates at

each split, $mtry \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ (in the software, the default regression value is one-third of the input), and the number of trees to grow, $ntree \in \{300, 500, 1000, 2000, 5000\}$. According to the variable properties, setting more regression trees can increase the stability of the estimated variable results (diaz-uribe and DE Andres, 2006); thus, we set $ntree = 500$. The default value of $mtry$ is one third of the total number of predictors, and there are six input variables for prediction, so $mtry = 2$.

2.5.2. Support vector machine (SVM)

The SVM regression algorithm is a nonlinear model based on statistical learning theory (Vapnik, 2000). The SVM algorithm uses effective kernel functions to project data into the high-dimensional feature space and then performs a simple linear regression in the high-dimensional space (Hornik et al., 2006). The resulting linear regression function in the high-dimensional feature space corresponds to a nonlinear regression in the original input space (Smola and Schölkopf, 2004). Due to its applicability and robustness, this algorithm is widely used in regression and classification problems (Xu et al., 2018; Chen and Ge, 2019). When using an SVM algorithm for variable prediction, the selection of the kernel function and parameter C is critical for achieving satisfactory prediction accuracy. When training the SVM model, we tested different values of the tuning parameters. We selected a radial basis function (RBF) as the kernel function, and the optimal model parameter C that minimizes the root-mean-square error (RMSE) of the prediction was determined.

2.5.3. Back propagation neural network (BPNN)

Through forward and backward propagation, BPNN continuously distributes the error to each layer of “neurons” until the error of the neural network reaches an acceptable range, thus achieving the regression of the dependent variable. The BPNN algorithm has a high self-learning potential and broad applicability, which makes it an effective method for solving nonlinear problems (Daniel et al., 2003; Wang et al., 2018). When building the BPNN prediction model, we selected *Tan-sigmoid* and *purelin* as the transfer functions of the hidden and output layers, respectively; the number of nodes in the hidden and output layers were chosen as 12 and 1, respectively, and *trainlm* was selected as the training function.

2.6. Model evaluation

The 315 samples were divided at a rate of 2:1 into a calibration set and a validation set. The coefficient of determination (R^2), RMSE, and residual predictive deviation (RPD) were calculated as indexes to evaluate the performance of the different models (Williams and Norris, 2001). The corresponding formulas are given below:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

$$RPD = \frac{SD}{RMSE} \quad (9)$$

where n is the number of samples, y_i is the observed SOC value of sample i , $\hat{y}_i$ is the predicted SOC of sample i , and SD is the standard deviation of the predicted SOC value. In general, a well-performing model usually has high R^2 and RPD values and a low RMSE (Williams and Norris, 2001). We referred to the following performance criteria: $RPD \geq 2$ reflects an excellent prediction, $1.4 \leq RPD < 2$ is good and $RPD < 1.4$ is poor (Chang et al., 2001).

Table 2

Statistical descriptions of the SOC contents in the whole, calibration and validation datasets.

Set	N	Max (g kg ⁻¹)	Min (g kg ⁻¹)	Mean (g kg ⁻¹)	SD (g kg ⁻¹)	CV ^b (%)
Whole dataset	315	45.84	2.47	22.46	6.87	30.57
Calibration dataset	210	45.84	2.47	22.45	6.86	30.54
Validation dataset	105	45.83	3.72	22.48	6.89	30.64

Note: a represents the SD, where a larger value indicates a more discrete and more unstable data set, and the smaller the SD is, the smaller the degree of dispersion of the data set and the closer the values are to the average; b denotes the CV, the meaning of which is consistent with the SD, but it eliminates the influence of the number of dimensions.

3. Results

3.1. Descriptive statistics of the SOC content

The descriptive statistics of the SOC content are presented in Table 2. The SOC content of the whole set of samples showed a wide range (2.47–45.84 g/kg) with a mean of 22.46 g/kg, an SD of 6.87 g/kg and a coefficient of variation (CV) of 30.57 %. The high SD and CV reflect the strong spatial variability of SOC.

3.2. Characteristics of soil spectral reflectance associated with the SOC content

The soil reflectance spectra of different SOC contents are shown in Fig. 3a. The spectral reflectance decreased with increasing SOC content, and the reflectance spectra show similar spectral shapes. Due to the presence of organic matter and iron ions, the reflectance increased rapidly with increasing wavelength between 433 and 1342 nm. There appears to be a reflection peak at 900–1050 nm caused by noise at the junction of the two sensors. The reflectance decreased rapidly with increasing wavelength from 1460–1600 nm; between 1600 and 1763 nm, the reflectance increased with increasing wavelength; between 1990 and 2445 nm, the reflectance increased and then decreased

with the wavelength. The spectrum curve is basically consistent with the laboratory spectrometer curve and the airborne hyperspectral curve.

3.3. Soil spectral characteristics after discrete wavelet transform

3.3.1. Reflectance

The spectral reflectance curves at different DWT decomposition scales are shown in Fig. 3b–k. At the L₁–L₃ scales, DWT reduced the level of spectral noise, thereby smoothing the spectral curve, especially in the 2200–2445 nm region. For example, in L₁, there exists a small burr in 2200–2445 nm due to the transmission of noise in the OR. In L₂ and L₃, the high-frequency signal is further removed; hence, this noise transmission phenomenon becomes increasingly weak. Consequently, only a small amount of noise is present in L₄–L₁₀. With further DWT decomposition, detailed spectral information is continuously removed, leading to spectral smoothing. In L₄, the DWT method eliminates the absorption valley at 1200–1300 nm, thus reducing the difference in spectral characteristics among different SOC contents. In L₅, the absorption peak at 900–1050 nm is eliminated. In L₆–L₁₀, DWT eliminates most of the detailed information in the spectral curve; therefore, there were no obvious differences in the shapes of the soil spectral curves with different SOC contents.

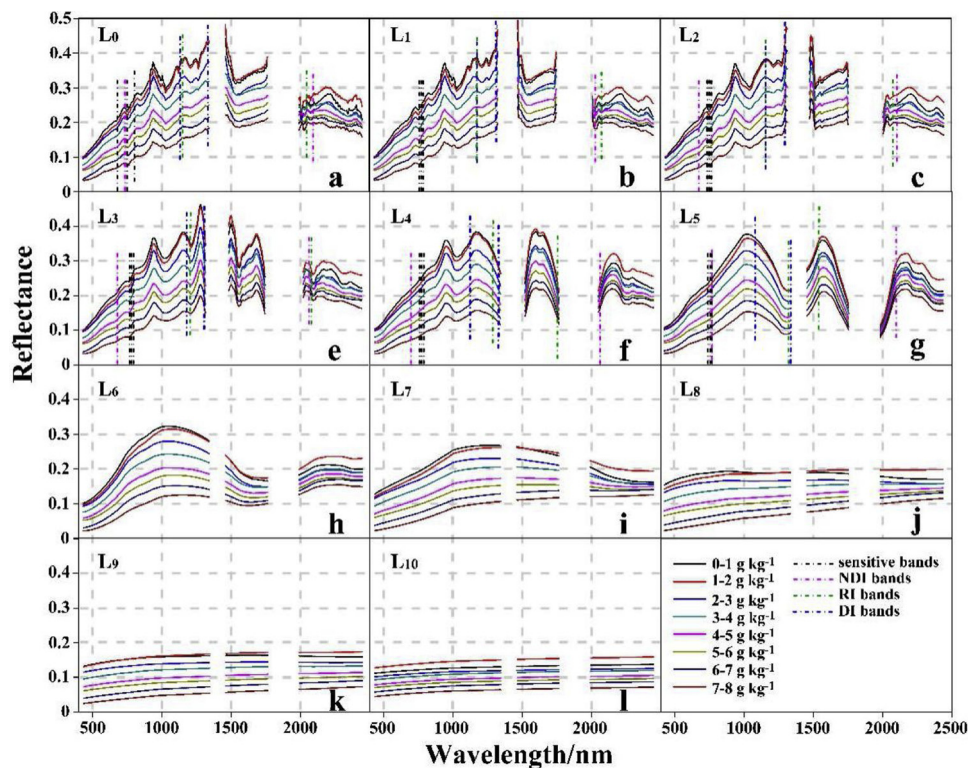


Fig. 3. Comparison of the differences among the soil reflectance curves with different SOC contents at different DWT decomposition scales.

Note: L₀ represents the original soil spectral reflectance curve; L₁–L₁₀ represent the reconstructed spectral reflectance curves based on DWT at decomposition scales of 1–10.

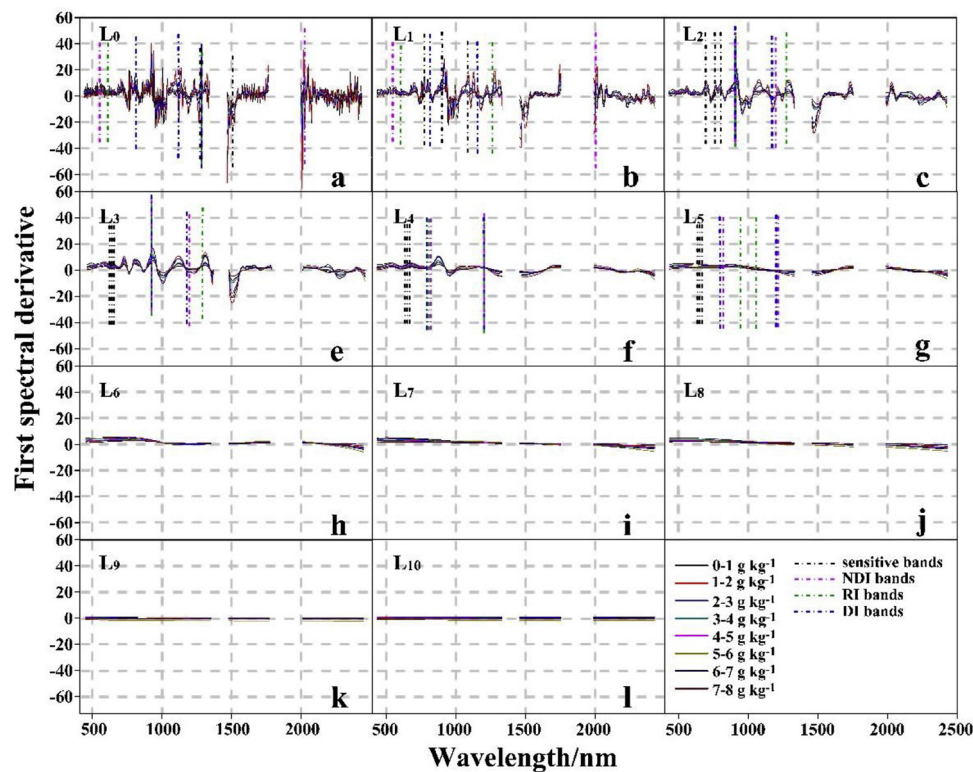


Fig. 4. Differences in the FDR curves with different SOC contents at different DWT decomposition scales.

3.3.2. FDR

In the OR (Fig. 3a), the absorption features of different components were impacted by other components to varying extents. The FDR can discriminate absorption peaks as well as enhance weak absorption features (Wang et al., 2018). However, the FDR is sensitive to noise (Fig. 4a); thus, we applied DWT with FDR to reduce the noise level.

Fig. 4 shows the FDR curves at different DWT decomposition scales. In L_0 – L_4 , DWT mainly eliminated the high-frequency component (noise) of the FDR; thus, the absorption characteristics at each position in the FDR curves were well preserved, but the shape of the absorption peak became more rounded, and for L_4 , the noise in the spectral data was minimal. For L_5 – L_{10} , with an increase in the decomposition scale, the low-frequency component (spectral characteristics) of the data was further eliminated, and the spectral curve tended to become smooth, resulting in the disappearance of absorption peaks at different positions. Especially for L_6 and L_{10} , the FDR absorption characteristics with different SOC contents are almost the same, hence, in this paper, the maximum decomposition scale is determined to be 5, and a further analysis is made on the basis of L_0 – L_5 .

3.4. Selection of predictor variables

We used PCA to calculate the W of each band and select the 3 bands with the highest W values and the 3 spectral indexes with the highest SOC correlation as the inputs.

3.4.1. Selection of spectral reflectance

We selected 3 optimal bands with the maximum W values as the inputs for the SOC prediction model to reduce both the redundancy of hyperspectral data and the complexity of the prediction model. This approach not only maintains the same number of inputs at different DWT decomposition scales but also enhances the comparison among various processing forms. The extracted optimal bands, main scores and W values are shown in Table 3. In OR, the optimal bands of different DWT decomposition scales are mainly concentrated in the range of

700–800 nm because the slope associated with this band width increases with increasing SOC content, and for L_3 the R values between the optimal bands and SOC is the highest. For the FDR, as the DWT decomposition scale increases, the selected optimal bands gradually move towards the visible range, mainly because DWT obviously eliminates the spectral characteristics above 1300 nm. For example, the optimal band is approximately 1300 nm for L_0 , whereas for L_5 , the optimal band is mainly approximately 630 nm. When the decomposition scale is L_2 , the R value is the highest between the optimal bands and SOC.

3.4.2. Selection of spectral indexes

The correlations between the NDI, RI, and DI and SOC were analysed by the contour map method (Fig. 5). In OR, the highly correlated region between the NDI and SOC is near the combination of the 1950 nm and 650 nm bands, and the highly correlated region between the RI and SOC is near the combination of the 2000 nm and 1150 nm bands. The high correlation between the DI and SOC is observed mainly in the absorption region of organic matter and iron ions (the combination of the 1100 nm and 1350 nm bands), which is consistent with the research results of Jin et al. (2016). There are large differences in the highly correlated region between the NDI and SOC at different DWT decomposition scales based on FDR (Fig. 6). For L_0 – L_1 , the highly correlated region is near the combination of the 2000 nm and 545 nm bands; however, for L_2 – L_5 , the highly correlated region is near the combination of the 1200 nm and 850 nm bands. The highly correlated region between the RI and SOC is near the combination of the 1250 nm and 800 nm bands, and the highly correlated region between the DI and SOC is near the combination of the 1200 nm and 800 nm bands. In other words, for each spectral index, the selected bands are mainly concentrated in the range of 500–1300 nm, which represents a region of remarkably high correlation between each band and SOC (Brown et al., 2006; Daniel et al., 2004; Liu et al., 2007a,b; Liu et al., 2009a,b); thus, the optimal spectral index selection result is reasonable.

Table 4 shows the correlations between the spectral indexes and

Table 3

Selected optimal bands determined by principal component analyses at different DWT decomposition scales.

Type ^a	Scale	Band ₁	Scores ^b	W _n ^c	R ^d	Band ₂	Scores	W _n	R	Band ₃	Scores	W _n	R	Rate ^e
OR	L ₀	766	0.97	0.04	−0.56**	809	0.97	0.03	−0.54**	690	0.96	0.03	−0.55**	98.59 %
	L ₁	766	0.98	0.03	−0.56**	762	0.97	0.03	−0.55**	771	0.94	0.02	−0.56**	98.86 %
	L ₂	762	0.97	0.04	−0.57**	766	0.95	0.04	−0.57**	758	0.95	0.03	−0.55**	97.95 %
	L ₃	775	0.94	0.05	−0.58**	771	0.95	0.05	−0.56**	779	0.87	0.05	−0.57**	99.07 %
	L ₄	775	0.92	0.06	−0.55**	771	0.93	0.05	−0.55**	766	0.86	0.05	−0.54**	99.29 %
FDR	L ₅	753	0.93	0.05	−0.53**	758	0.89	0.05	−0.55**	762	0.82	0.04	−0.54**	99.55 %
	L ₀	1502	0.95	0.03	0.56**	1274	0.97	0.03	−0.56**	1266	0.91	0.03	−0.55**	87.71 %
	L ₁	899	0.98	0.04	−0.61**	771	0.94	0.04	−0.58**	1080	0.96	0.03	−0.60**	90.60 %
	L ₂	758	0.92	0.05	−0.64**	792	0.92	0.05	−0.62**	689	0.90	0.04	−0.57**	94.39 %
	L ₃	621	0.90	0.05	−0.59**	625	0.93	0.05	−0.60**	617	0.91	0.05	−0.63**	96.69 %
	L ₄	655	0.95	0.04	−0.60**	659	0.94	0.03	−0.58**	651	0.95	0.03	−0.62**	99.22 %
	L ₅	630	0.96	0.04	−0.56**	625	0.96	0.04	−0.54**	634	0.96	0.03	−0.61**	99.55 %

Note: a represents the type of input; b represents the main score of the band; c represents the weight of each band; d represents the correlation coefficient between the band and SOC; e represents the cumulative contribution rate of the band and ** represents significance at the $p < 0.01$ level.

SOC at different decomposition scales. In OR, the correlations between the NDI, RI, and DI and SOC barely change for L₀–L₂; however, for L₃–L₅, these correlations begin to decline. For the FDR, with an increase in the decomposition scale, the correlations between the spectral indexes and SOC tend to increase initially and then decrease, and for L₂, the correlations between the spectral indexes and SOC are the highest. Specifically, the correlations between SOC and the NDI, RI and DI are computed as $(R_{1165}-R_{905})/(R_{1165}+R_{905})$, R_{1265}/R_{905} and $R_{1160}-R_{905}$, respectively, with R values of -0.69, 0.67 and -0.73, respectively.

3.5. The results of different SOC prediction models

3.5.1 SOC prediction based on RF

The SOC prediction accuracies using the OR bands with different DWT decomposition scales as inputs followed the order of $L_3 > L_1 > L_4 > L_2 > L_0 > L_5$ (Fig. 7). The R^2 values of L₁–L₄ are all greater than 0.70, the RMSEs are all less than 3.80 g kg^{−1}, and only the RPDs of L₃ and L₄ are greater than 1.4. An SOC prediction model was constructed at different DWT decomposition scales with the FDR as the input; the prediction accuracy initially increased and then decreased as the scale of DWT increased and reached the highest accuracy at the L₂ scale ($R^2 = 0.79$, RMSE = 3.63 g/kg^{−1}, RPD = 1.46). When the decomposition scale was L₂–L₄, the RMSE was low (less than 3.7), and the RPD was greater than 1.4, which indicates that the L₂–L₄ scales can provide a better SOC prediction than other scales on the basis of the FDR.

3.5.1. SOC prediction model based on SVM

The SOC prediction accuracies at different DWT decomposition scales were $L_2 > L_1 > L_0 > L_3 > L_5 > L_4$ when using the OR as the input, which resulted in the lowest RMSE (3.89 g kg^{−1}) and the highest R^2 (0.71) and RPD (1.17) for L₂. However, as shown in Fig. 7, all of the RPDs at different decomposition scales were less than 1.4, illustrating that the prediction performance of the OR combined with an SVM prediction model is poor. When predicted based on the FDR, the SOC prediction accuracy gradually increases to a maximum value and then decreases. At the L₂ decomposition scale, the SVM prediction algorithm achieved a high accuracy (R^2 : 0.80, RMSE: 3.20 g kg^{−1}, RPD: 1.72), much higher than that with the OR inputs ($R^2 = 0.71$). An RPD value > 1.4 suggests that the FDR combined with an SVM model leads to accurate SOC content predictions.

3.5.2. SOC prediction model based on BPNN

According to Figs. 7 and 8, this study found that the predicted SOC content of this model was the closest to the measured SOC content, and this high-accuracy model had the highest R^2 of 0.83 in this study. Among the various prediction models, the different types of inputs with DWT showed considerable differences in the improvement of the SOC

prediction accuracy. For the OR, the R^2 of the optimal decomposition scale increased only by 3% compared with the reflectance at the L₀ decomposition scale, the RMSE decreased by approximately 0.30 g kg^{−1}, and the RPD increased by 0.17. For the FDR, the R^2 and RPD values of the optimal prediction model were improved by approximately 9% and 0.60, respectively, and the RMSE decreased by 0.85 g kg^{−1}, making this model highly stable. Therefore, compared with the method in which the OR is combined with DWT, the method in which the FDR is combined with DWT is better at improving the SOC prediction accuracy.

Fig. 8 displays a scatter plot of different SOC prediction models with the optimal FDR decomposition scale. This figure shows that the predictive performance of the different prediction models follows the order of BPNN > SVM > RF, but the difference in accuracy among these prediction models is small, which suggests that the input selection process is objective and accurate. In the RF and SVM models, the predicted values exhibit a larger deviation from the 1:1 line at higher SOC levels than do those for other models. In contrast, the predicted values of the BPNN model are close to the 1:1 line at all SOC contents, which confirms the strong predictive power of the BPNN model.

3.6. Mapping of SOC using GF-5 hyperspectral satellite data

To demonstrate the results of different prediction models, the mapping results of the SOC content derived from GF-5 images are shown in Fig. 9. The mapping results are obtained by different prediction models using the FDR with the optimal DWT decomposition scale. For example, Fig. 9a is the mapping result obtained by the RF prediction model using the FDR, and the optimal decomposition scale is 2. Similarly, Fig. 9b is the mapping result obtained with the SVM and Fig. 9c is the mapping result based on the BPNN, which is also based on previously verified results, confirming that the optimal DWT decomposition scale based on the FDR is 2. The spatial distributions of SOC derived using different prediction models are similar. The highest SOC content was observed in the northeastern part of the study area, which is mainly composed of Phaeozems and Cambisols. The high-latitude frigid environment has a large fixed amount of humus. Compared with other areas, forest conditions in this area limit the duration and intensity of reclamation. The southwestern region is an alluvial plain, mostly composed of Chernozems and Arenosols, with relatively large spans of pastoral areas; however, soil erosion and land desertification herein are caused by high river water flows and groundwater levels, and thus, soil alkalization is quite serious. Therefore, SOC shows a gradual decrease from the northeast to the southwest, and the spatial distribution of SOC in the study area significantly varies.

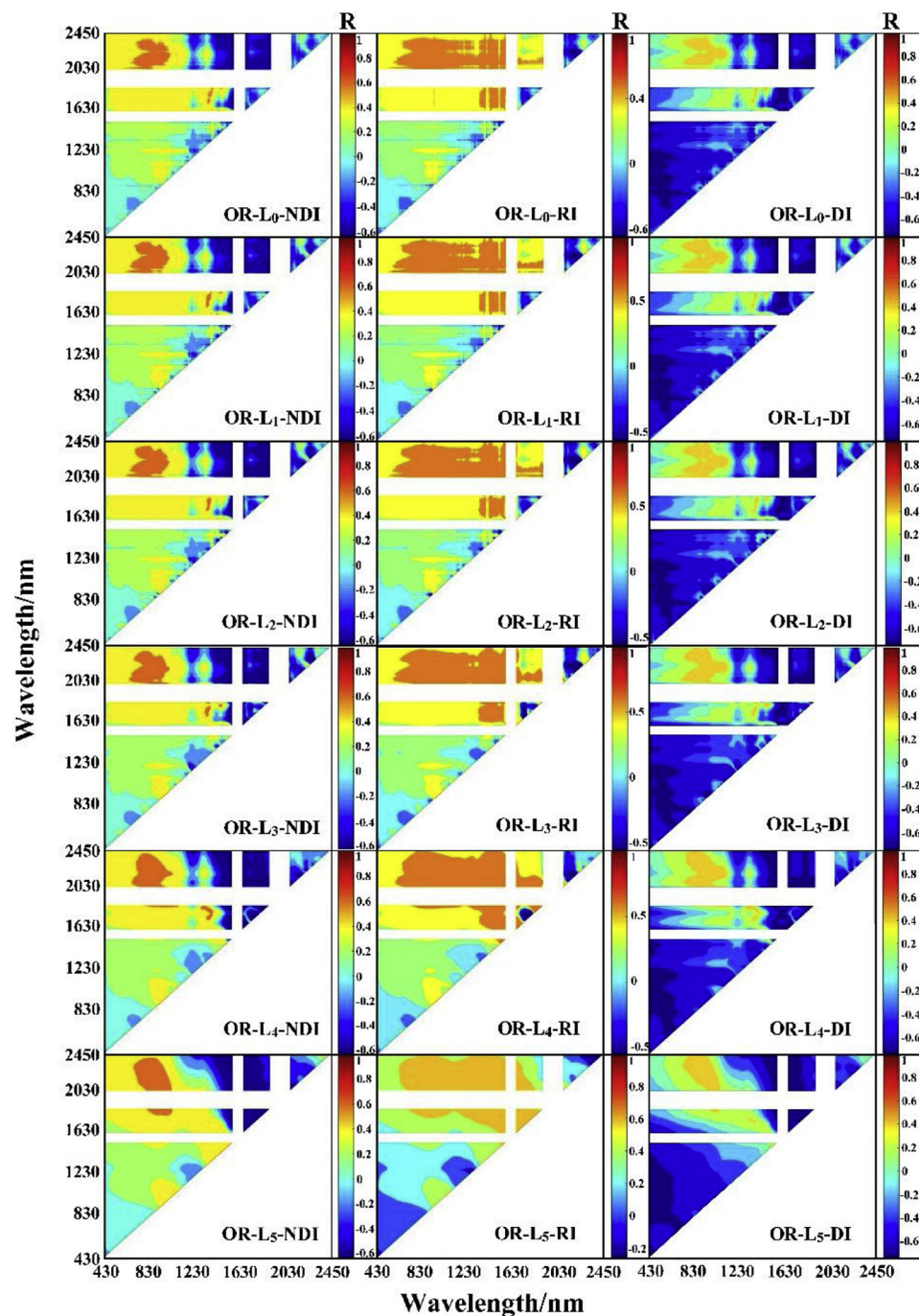


Fig. 5. Correlations between the optimal spectral indexes and the soil organic carbon content at different reflectance decomposition scales. Note: OR-L₀-NDI represents the correlation between the NDI and SOC for reflectance data not decomposed by DWT.

4. Discussion

4.1. Advantages of DWT for reducing noise in hyperspectral satellite data

In most studies using DWT, the data were decomposed into different scales; however, with the increase in scale, the dimension of the data decreases, which makes it difficult to compare the wavelet coefficients among different scales (Blackburn and Ferwerda, 2008). Thus, we reconstructed the wavelet coefficients at different scales to enable a comparison and to evaluate the optimal decomposition scale; this approach may be useful to show how DWT can be used to reduce noise effectively and select the best scale for hyperspectral data. Compared with other noise reduction methods, DWT can perform spectral analysis

of local signals, giving DWT more advantages in the prediction of soil properties than other noise reduction methods (Biswas, 2018). The noise reduction effect of DWT is very obvious. As shown in Fig. 3a, the noise between 2200 and 2450 nm is relatively large, which results in the appearance of small burrs in the spectral curve. With an increase in the decomposition scale, the noise in the spectral curve diminishes, so the curve becomes smooth. Moreover, in the DWT decomposition process, the spectral characteristics of the soil are retained, especially some large absorption valleys and peaks (Figs. 3 and 4) (Kaewpijit et al., 2002).

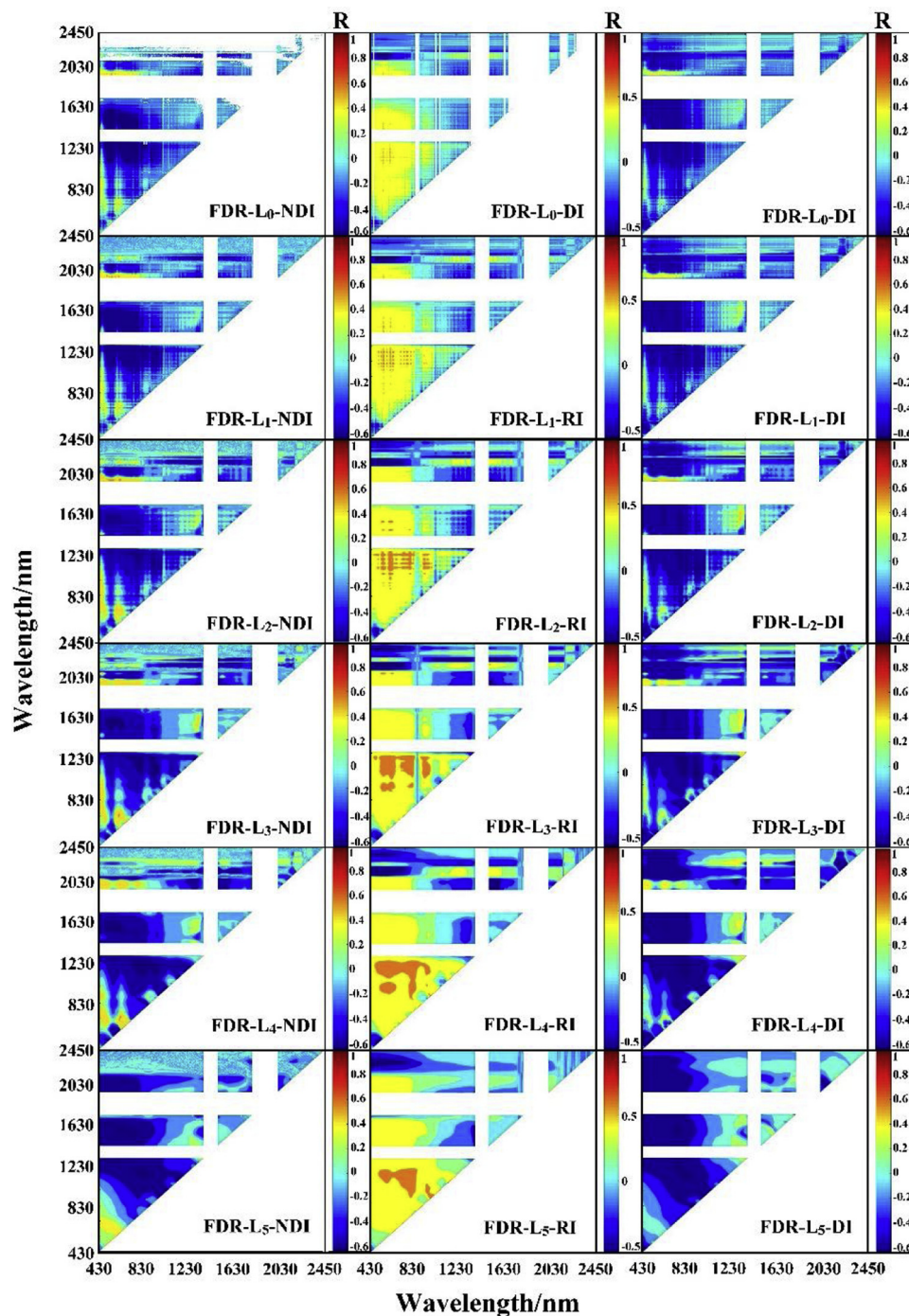


Fig. 6. Correlations between the optimal spectral indexes and the soil organic carbon content at different scales of first- derivative spectral reflectance decomposition. Note: FDR- L_0 -NDI represents the correlation between the NDI and SOC for the first-derivative reflectance data not decomposed by DWT.

4.2. Advantages of the DWT for FDR

Compared with the combination of the OR with DWT, the combination of the FDR with DWT has a more obvious effect on improving the SOC prediction accuracy. This is mainly because when the OR is combined with DWT analysis (Fig. 10), the characteristics of different decomposition scales minimally change. For example, for L_0 - L_3 , as the decomposition scale increases, noise is gradually removed, resulting in a smoother curve; however, the shapes of the two absorption peaks at 1050–1200 nm and 1500–1600 nm change little (Fig. 10a and b). When the FDR is combined with DWT analysis (Fig. 10), as the decomposition scale increases, the spectral characteristics of different

decomposition scales vary greatly. For L_4 , the spectral curve loses almost all of its spectral characteristics. However, the absorption peaks and valleys of the spectral curve gradually become smooth and more regular as the decomposition scale increases, which is the main reason why the correlation between the FDR and SOC is improved.

4.3. Physical meaning of the selected predictor variables

Previous findings suggested that full spectra or the maximum amount of available spectral data will require an increased computation time and that it is almost impossible to improve the prediction accuracy (Darvishzadeh et al., 2008). Therefore, this paper extracted 6 inputs

Table 4

Correlation coefficients (R) between the spectral indexes and SOC.

Type	Scale	NDI	R	RI	R	DI	R
OR	L ₀	(R ₂₁₃₄ -R ₇₆₂)/(R ₂₁₃₄ +R ₇₆₂)	-0.66**	R ₂₀₇₄ /R ₁₁₆₄	0.68**	R ₁₁₅₆ -R ₁₃₄₂	-0.76**
	L ₁	(R ₂₀₁₅ -R ₆₈₁)/(R ₂₀₁₅ +R ₆₈₁)	0.65**	R ₂₀₅₈ /R ₁₁₆₄	0.68**	R ₁₁₆₄ -R ₁₃₄₂	-0.76**
	L ₂	(R ₂₀₇₄ -R ₆₈₉)/(R ₂₀₇₄ +R ₆₈₉)	0.65**	R ₂₀₇₄ /R ₁₁₆₄	0.68**	R ₁₁₆₄ -R ₁₃₄₂	-0.76**
	L ₃	(R ₂₀₄₉ -R ₆₇₂)/(R ₂₀₄₉ +R ₆₇₂)	0.65**	R ₂₀₅₈ /R ₁₁₉₈	0.67**	R ₁₁₇₄ -R ₁₃₄₂	-0.75**
	L ₄	(R ₂₀₅₈ -R ₆₉₈)/(R ₂₀₅₈ +R ₆₉₈)	0.64**	R ₁₇₆₃ /R ₁₂₈₃	0.66**	R ₁₁₂₅ -R ₁₃₄₂	-0.75**
	L ₅	(R ₂₀₉₁ -R ₇₆₂)/(R ₂₀₉₁ +R ₇₆₂)	0.64**	R ₁₅₃₆ /R ₁₃₁₆	0.66**	R ₁₀₈₀ -R ₁₃₄₂	-0.74**
FDR	L ₀	(R ₁₉₉₉ -R ₅₄₄)/(R ₁₉₉₉ +R ₅₄₄)	0.64**	R ₁₂₆₆ /R ₆₀₀	0.65**	R ₁₂₆₆ -R ₈₁₀	-0.71**
	L ₁	(R ₁₉₉₉ -R ₅₄₄)/(R ₁₉₉₉ +R ₅₄₄)	0.65**	R ₁₂₅₈ /R ₆₀₀	0.66**	R ₁₁₄₈ -R ₈₁₀	-0.71**
	L ₂	(R ₁₁₆₅ -R ₉₀₄)/(R ₁₁₆₅ +R ₉₀₄)	-0.69**	R ₁₂₆₆ /R ₉₀₄	0.67**	R ₁₁₅₆ -R ₉₀₄	-0.73**
	L ₃	(R ₁₁₇₃ -R ₉₀₄)/(R ₁₁₇₃ +R ₉₀₄)	-0.68**	R ₁₂₇₅ /R ₉₀₄	0.66**	R ₁₁₂₃ -R ₉₀₄	-0.72**
	L ₄	(R ₁₁₉₉ -R ₈₁₄)/(R ₁₁₉₉ +R ₈₁₄)	-0.66**	R ₁₁₉₉ /R ₈₀₁	0.66**	R ₁₁₉₉ -R ₇₉₇	-0.71**
	L ₅	(R ₁₁₉₀ -R ₈₀₁)/(R ₁₁₉₀ +R ₈₀₁)	-0.64**	R ₁₀₃₈ /R ₉₂₅	0.64**	R ₁₁₉₂ -R ₇₈₄	-0.70**

Note: ** represents significance at the $p < 0.01$ level.

that have strong correlation with the SOC content, including 3 bands and 3 spectral indexes, namely, the NDI, RI and DI (Jin et al., 2016). Compared with other studies that used only a single indicator, this experiment used multiple indicators with clear physical meaning as inputs, thereby avoiding data redundancy and ensuring the integrity of the information. The sensitive bands of OR with different decomposition scales were concentrated between 700 and 800 nm (Fig. 3, Table 3), which is mainly due to the presence of iron ions and organic matter. For the FDR, the sensitive bands are 1502, 1274, and 1266 nm when the decomposition scale is L₀, mainly due to the overtones and combinations of fundamental vibrations associated with the stretching and bending of dominant OH, CH, and NH groups, resulting in a direct spectral response to C in the VNIR range (Clark et al., 1990; Stenberg et al., 2010); additionally, FDR processing increases the correlation between these bands and SOC at the edges of the VNIR range. This result is consistent with the finding of a previous study by Lu et al. (2007). With an increase in the decomposition scale, the bands sensitive to SOC gradually shift towards the visible range (Table 3) because the spectral response band of SOC is concentrated mainly in the range

of < 1000 nm and the absorption characteristics of the spectral response wavelength range without SOC (> 1300 nm) are effectively eliminated (Fig. 10c); moreover, the spectral difference between different SOC contents decreases (Fig. 4), and the sensitivity between the SOC content and spectral curve declines, which further validates the spectral response band position of the SOC in the previous study. The sensitive bands selected in this paper are not located in the noisy area mentioned in Section 3.2 (900–1050 nm). Compared with the sensitive bands, the selected optimal NDI, RI and DI indicators have a higher correlation with SOC, which is consistent with the results of previous research (Liu et al., 2007a, b; He et al., 2009; Jin et al., 2016). We used FDR combined with DWT, and when the optimal decomposition scale was 2, the SOC prediction accuracy reached the maximum with the BPNN model. The 3 selected optimal bands are 758 nm, 792 nm, and 689 nm, and the 3 spectral indexes are (R₁₁₆₅-R₉₀₅)/(R₁₁₆₅+R₉₀₅) (NDI), R₁₂₆₅/R₉₀₅ (RI), and R₁₁₆₀-R₉₀₅ (DI); the selected bands are all within the range of the SOC spectral response bands. The bands at 758, 792, 689 and 1265 nm correspond to the highest value of the FDR absorption peak; with an increase in the SOC content, this value

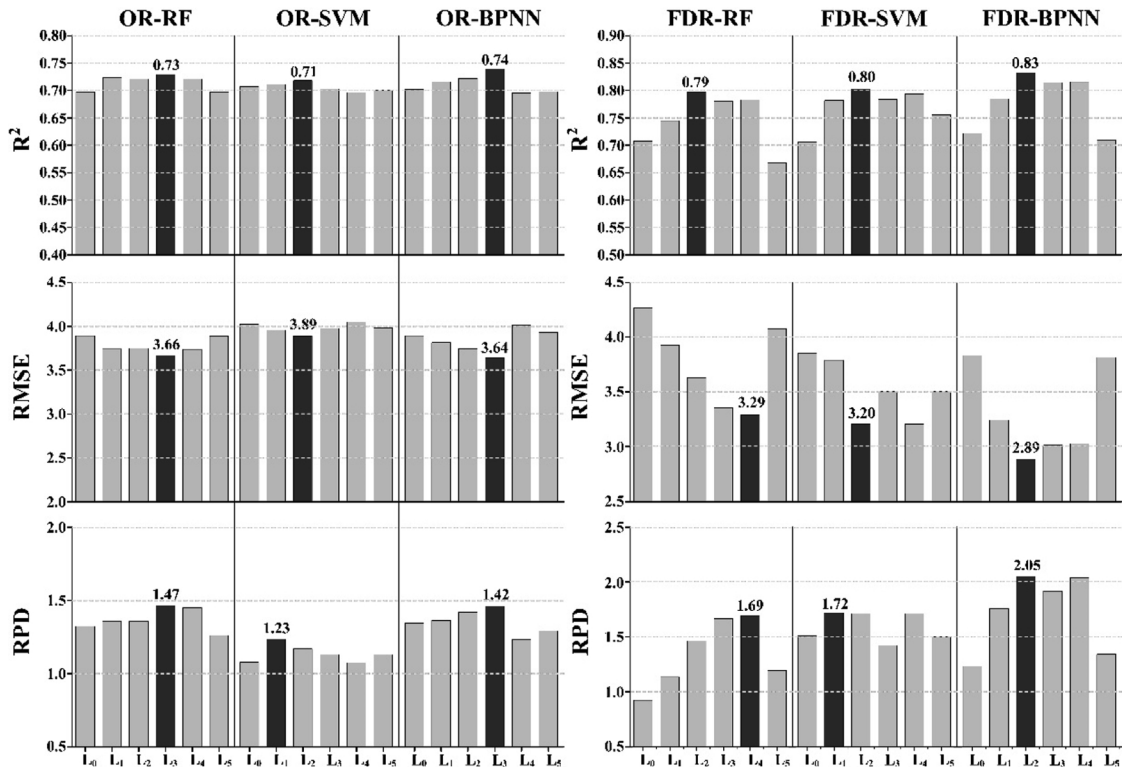


Fig. 7. SOC prediction accuracy at each decomposition scale based on different prediction models.

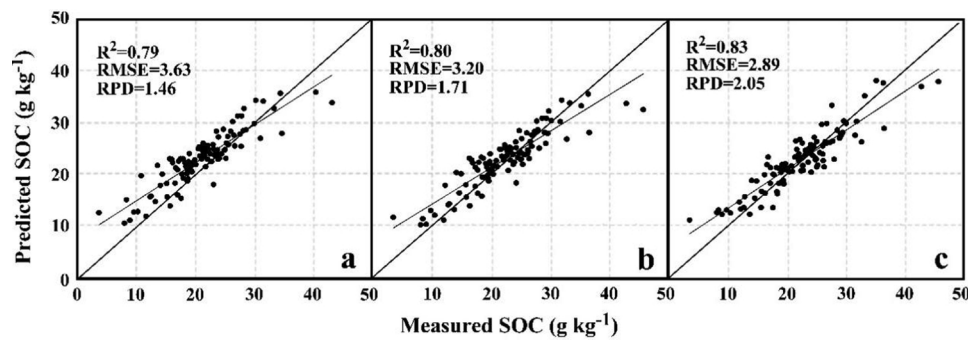


Fig. 8. Comparison of measured vs. predicted SOC contents using the optimal decomposition scale based on different FDR-DWT prediction models. Note: a represents the RF model; b represents the SVM model and c represents the BPNN model.

decreases, and the shape of the absorption peak becomes smoother. The bands at 905, 1160, 1165 and 1265 nm correspond to the highest value of the FDR absorption valley; with an increase in the SOC content, this value becomes higher and the shape of the absorption peak becomes smoother. Therefore, we can identify the specific bands that can improve the SOC prediction accuracy with hyperspectral satellite data, and we assert that the combination of band information and spectral information can reduce the computational burden when simulating soil environmental variables. Now, we seek to verify the advantage of the input selection method proposed for SOC prediction in this paper, that is, PCA is used to calculate the W of different bands to select the 3 bands with the highest W values and the 3 spectral indexes with the highest SOC correlations as the inputs. In this paper, PCA is used to transform the soil hyperspectral reflectance data into independent principal components, and a total of 18 principal components (contribution rate > 85 %) are extracted for SOC prediction; this method is commonly used to reduce the dimension of hyperspectral data. Compared with the accuracy of SOC prediction based on the input selection method proposed in this paper, the R^2 of this traditional method decreased by 8%, the RMSE improved by 3.04 g kg⁻¹, and RPD decreased by 0.78 (Table 5); hence, the input selection method proposed in this paper can potentially be widely used in future soil property research.

4.4. Comparison of different prediction models

A comparison of the RF, SVM and BPNN models shows that they all

have high prediction accuracy, but the RMSEs and RPDs vary considerably. The BPNN model has the highest accuracy with no apparent bias due to the high self-learning ability and applicability of the BPNN model (Wang et al., 2015). Thus, this model is very suitable for fitting data with nonlinear relations and complex interactions, and it can uncover the hidden relationships among inputs (Batchelor et al., 1997). We further discovered that when using DWT, the BPNN prediction accuracy is higher than the SVM prediction accuracy, which is consistent with the conclusions reached by Viscarra Rossel and Behrens (2010). Although BPNN is a “black box”, combining the selection of the optimal bands with nonlinear modelling helps achieve good predictability and interpretability. Compared with the BPNN model, the RF and SVM models have slightly lower prediction accuracy. The optimal model is not unique in the prediction of soil nutrients in different regions. When selecting the optimal prediction model, both the prediction accuracy of the model and the calculation time should be considered (Fyströ, 2002; Nawar et al., 2016). In this paper, the calculation time of the BPNN model is relatively long, which is mainly caused by the BPNN reducing the prediction error of SOC through the repeated gradient descent method. However, we do not consider the calculation time of the model as an evaluation index for SOC prediction. Therefore, the scope of this research should be expanded in future studies to consider the model calculation time and obtain a more universal SOC prediction model.

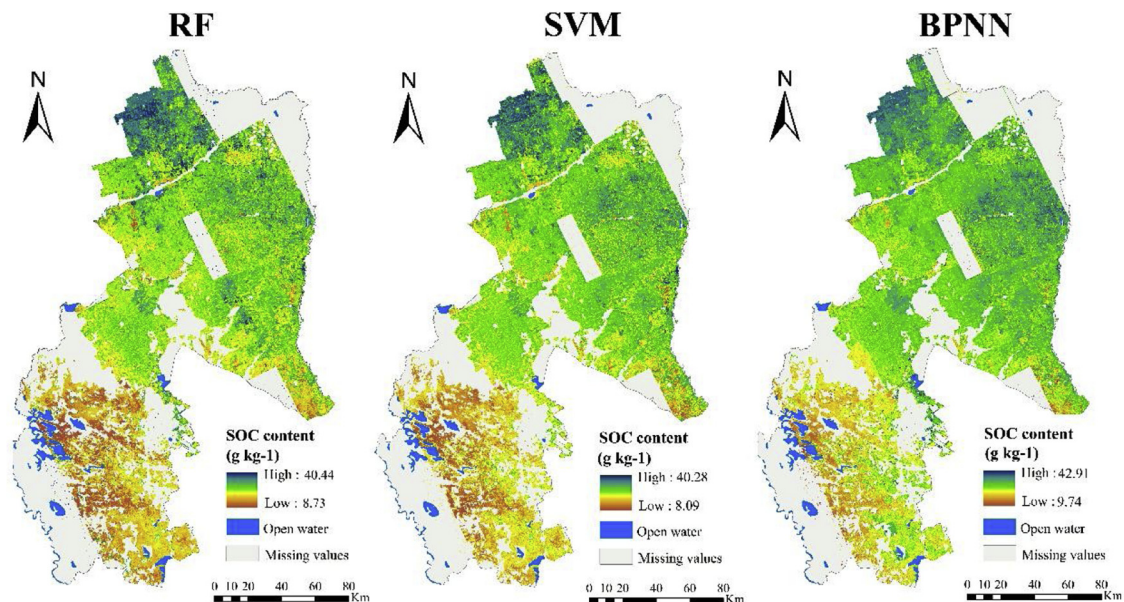


Fig. 9. SOC mapping results based on different methods.

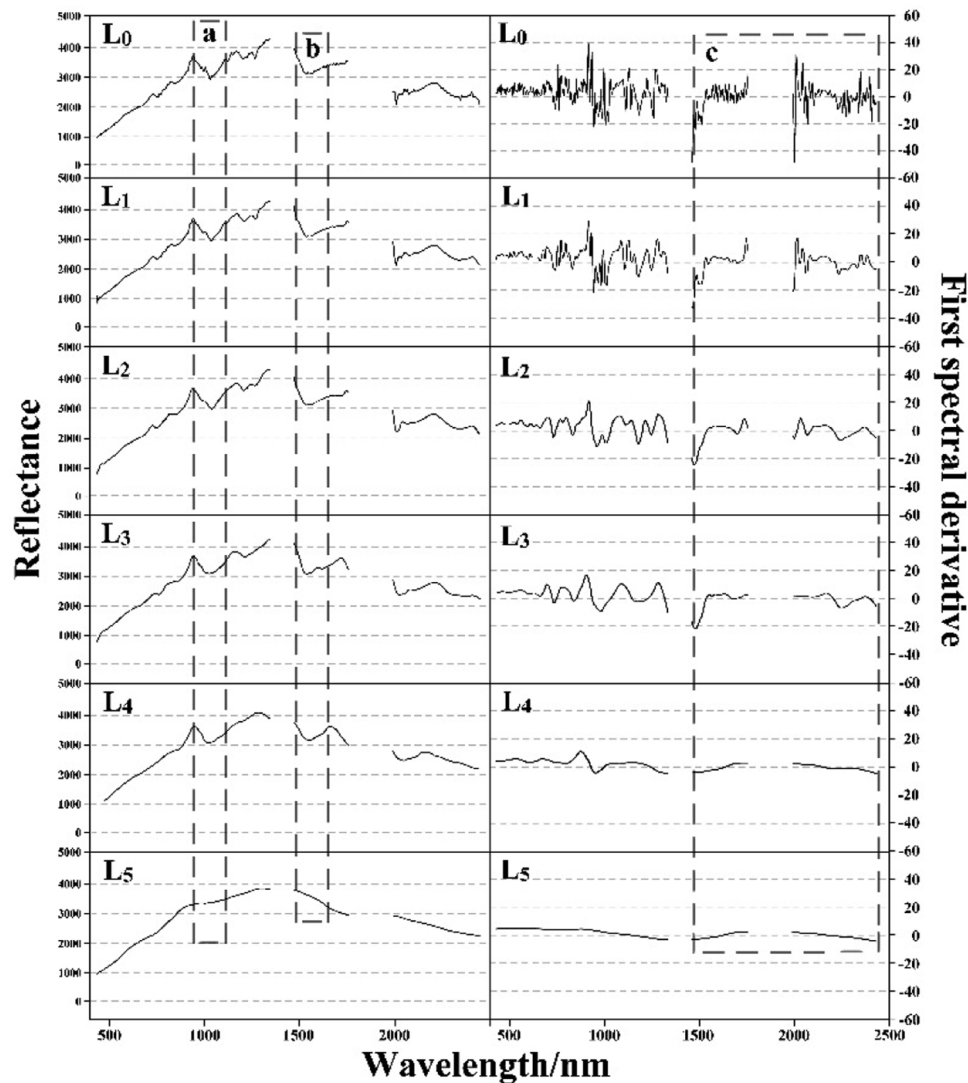


Fig. 10. The spectral characteristics of different decomposition scales based on reflectance and the first-spectral derivative.

Table 5

Comparison of the SOC prediction results of different input selection methods.

Type	Scale	Inputs	Prediction model	R ²	RMSE	RPD
FDR	2	689, 758, 792, NDI, RI, DI	BPNN	0.83	2.89	2.05
FDR	2	18 PCs	BPNN	0.75	5.93	1.27

Note: PCs represent the transformed independent principal components.

4.5. Advantages of hyperspectral satellite data for SOC prediction

Hyperspectral remote sensing technology is extensively used to obtain spectral information in great detail with the aim of improving the soil property prediction accuracy; the significant advantage of this approach lies in the enormous amount of band information that can be acquired in comparison with the abilities of multispectral satellites. Taking the Landsat OLI sensor as an example, an entire OLI image contains a total of 9 bands, and thus, limiting any information may cause some prediction-related details to be ignored (Fig. 11); furthermore, the wavelength range correlated with the soil prediction is several times smaller than the 600–800 nm band, which has a high correlation with laboratory SOC predictions; thus, only 2/5 of the information is included, this band information is not sufficiently detailed, and unfortunately, different research results cannot be supported

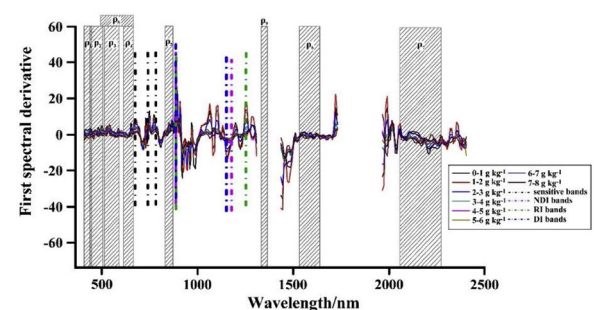


Fig. 11. Comparison based on the band range of Landsat OLI and the selected optimal bands of hyperspectral data.

Note: p_1 – p_9 represent band 1–band 9 of Landsat OLI; the curves represent the first-derivative reflectance of the hyperspectral data.

or verified with this lack of band information (Brown et al., 2006; Dalal and Henry, 1986; Daniel et al., 2004; Hummel et al., 2001; Kweon and Maxton, 2013; Liu et al., 2009a,b; Palacios-Orueta and Ustin, 1998; Rossel et al., 2006; Shepherd and Walsh, 2002; Sudduth and Hummel, 1991). In the model with the highest prediction accuracy, most of the selected bands are not in the spectral range of multispectral satellites, and the spectral differences of different SOC contents are not obvious in the multispectral data (Fig. 11). Therefore, we believe that

hyperspectral satellite data have more advantages than multispectral satellite data in SOC prediction. However, the SNR of a hyperspectral sensor is typically lower than that of a multispectral sensor; this discrepancy is due to the reduction in energy harvested by a hyperspectral sensor in a narrow band coupled with the low irradiance of the sun in the SWIR region, causing the SNR to decrease continuously. For example, Castaldi et al. (2014) obtained images acquired from the ALI and Hyperion sensors onboard the EO-1 satellite and combined those images with ground-sampling data to predict soil properties; the results showed that the Hyperion spectrometer does not have a significant predictive advantage due to its low SNR at the spectral characteristic wavelengths of clay minerals. In this paper, GF-5 hyperspectral data were used to invert the spatial distribution of SOC in the Songnen Plain region over an area of 4.25 billion ha. After removing noise from the hyperspectral data, the prediction accuracy based on the BPNN model was the highest ($R^2 = 0.83$, RMSE = 2.89 g kg^{-1} , RPD = 2.05). Compared with previous SOC prediction studies, the obtained prediction accuracy, which is directly related to the satellite data and processing scheme, is greatly improved. Clearly, the parameters (such as the spectral interval, number of bands, and SNR) of different spectral satellites vary, and the response curves for various substances are also different. Nevertheless, this paper verified the advantages of using satellite imagery to predict soil properties, i.e., by reducing noise and avoiding the effects of soil moisture under high SNR conditions, as noted by Castaldi et al. (2016). Additionally, information was obtained from remote sensing imagery at a specific position during a fixed acquisition period, and high-accuracy SOC mapping was performed at the regional scale.

In addition, when using satellite data to predict the SOC content, the influence of soil moisture on the SOC prediction accuracy cannot be ignored. The images obtained in March and April of 2019 were minimally affected by water factors because in these months, the Songnen Plain was relatively dry, and there was very little snowfall, resulting in the lack of snow cover and an arid surface. Thus, it can be concluded that soil moisture had almost no effect on the spectral reflectance of the surface soil. In this study, we imaged the bare soil to obtain hyperspectral satellite imagery. To invert the result and obtain the SOC content for a surface covered with straw or green vegetation, it is necessary to introduce environmental elements such as topography, and the specific method for achieving this goal must be further studied.

5. Conclusions

In this paper, GF-5 hyperspectral satellite data were used to predict the regional SOC content. A noise removal method for hyperspectral satellite soil data was proposed, DWT was used to decompose and reconstruct the OR and FDR at different scales, and a method was implemented to select the spectral inputs with objective, accurate and obvious physical meanings. The main results are as follows:

- (1) We built a robust SOC prediction model, and the SOC content in the northern Songnen Plain region was mapped, which will improve the estimation accuracy of the total carbon pool in terrestrial ecosystems.
- (2) The wavelet coefficients decomposed by DWT can be reconstructed and effectively compared with the wavelet coefficients at different decomposition scales. Moreover, the noise in hyperspectral satellite data can be eliminated at relatively low decomposition scales, and the SOC prediction accuracy can be effectively improved. However, with an increase in the decomposition scale, the differences in the spectral characteristics among different SOC contents will diminish, and the SOC prediction accuracy will decrease at a relatively high decomposition scale.
- (3) A method is implemented to select the 3 optimal inputs with the highest W values using PCA and the NDI, RI and DI indicators with the highest correlations with the SOC content by utilizing a contour

map. This approach eliminates redundant hyperspectral data and effectively retains the integrity of the hyperspectral data and the physical meaning of the original bands. In the model with the highest prediction accuracy, the selected bands are all within the range of the SOC spectral response bands, and the absorption valleys and peaks corresponding to these bands vary greatly with changes in the SOC content.

- (4) Hyperspectral satellite data can compensate for the band deficiency issues of multispectral satellite data, and the positions of the optimal inputs we selected are not within the range of the existing multispectral satellite bands; hence, the proposed approach greatly improves the SOC prediction accuracy.

These research results provide a new framework for predicting and mapping SOC contents over large regions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xiangtian Meng: Methodology, Software, Writing - original draft, Writing - review & editing, Formal analysis, Validation. **Yilin Bao:** Methodology, Software, Writing - review & editing, Formal analysis, Software. **Jiangui Liu:** Data curation, Writing - review & editing. **Huanjun Liu:** Conceptualization, Methodology, Funding acquisition. **Xinle Zhang:** Project administration. **Yu Zhang:** Resources, Investigation. **Peng Wang:** Resources, Investigation. **Haitao Tang:** Investigation. **Fanchang Kong:** Investigation.

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